On the discussion questions about Quine's 'On What There Is' (November 12, 2025)

As usual, here are a couple of thoughts and observations on the discussion questions for your perusal. Comments, questions, objections, additions: all are welcome!

All groups:
"Discuss in what way and to what extent Quine's criterion is connected with questions regarding reductionism."

As Quine himself also notes explicitly, it is not the criterion by itself that determines ontological commitments, it is something that is applied to a conceptual scheme and it is the conceptual scheme that carries the burden of deciding 'what there is'. A conceptual scheme is expressed in an interpreted language and is used to formulate a theory about an empirical domain. It is in that connection that the question about reductionism arises, i.e., it arises in the context of the *application* of the criterion

The criterion itself is neutral with respect to its application. Thus, if we can formulate the best scientific theory that we have at a given moment in time in a language that *requires* quantification over a particular type of entities, then we are committed to accepting their existence, in accordance with the criterion. For independent reasons, including a firm belief in the positive effects of applying Occam's razor ('Non sunt multiplicanda entia sine necessitate', i.e., one should not postulate more (types of) entities that one is forced to do), we may want to look for an alternative theory that has the same descriptive scope and the same explanatory power but that manages to do with fewer types of entities. So, the reduction of a certain type of entities to another type of entities is motivated primarily by considerations of *parsimony*: 'Less is more'. But note that, in the end, parsimony never trumps descriptive scope or explanatory adequacy.

So, in conclusion, Quine's criterion is neutral with respect to concrete questions about possible instances of reductionism: as such it does not take sides. However, in combination with independent additional criteria, such as ontological parsimony (Occam's razor), or an independently motivated preference for physicalistic over phenomenological accounts, or some, again, independently motivated, rejection of intensional entities, it can be put to work and help determine where reductionism is possible and, also important, at what cost.

Group 1:
"Does Quine's criterion favour a position in the fact-versus-'thing'-versus process debate? If so, which one? And does it show that other positions are wrong?"

It is clear from 'On What There Is' that when Quine discusses ontological commitments his focus is on entities: an ontological commitment is a commitment to the existence of entities. However, the nature of those entities to which a particular conceptual scheme can be shown to be committed using the criterion of ontological commitment is really a matter of the ontological scheme, not of the criterion of ontological commitment as such. It basically boils down to the choice of the domain over which the quantifiers in an interpreted language range. And this much we have seen already in the context of our introduction to predicate logical languages. Basic element of their interpretation, and hence of their use in representation and translation, is the choice of a domain. This supplies the entities that serve as interpretation of the individual constants and the relation symbols, and it is what the quantifiers 'there exists an x' and 'for all x' range over.

For Quine, the natural choice is to let the domain consist of macroscopic material objects, as these are what we can have direct sensory evidence of. From that perspective, entities that are not directly perceivable, but that we observe only with the help of instruments, or that we cannot perceive at all but that we (need to) postulate, come with theoretical assumptions, and thus need more, and theory-dependent, support. From that perspective a facts-ontology is more a construction than a direct given: we do not perceive facts in the same way we perceive objects. But the ontological commitment criterion itself is neutral at this point: we can postulate a domain of facts that our sentences refer to and let quantifiers range over them. The ontological commitment is then to facts, not to objects. And *mutatis mutandis* the same holds for processes. So, again we see that we need additional, and independently motivated, reasons for making a choice between a thing-, a fact-, or a process-based ontology. The ontological commitment criterion is neutral with respect to that choice; it only tells us what we are committed to once we made it.

Note that there can be a connection between facts-ontologies and things-ontologies. Atomistic fact-ontologies, such as Wittgenstein's (in the *Tractatus*) and Russell's (in *The Philosophy of Logical Atomism*), assume that there are 'atomic facts': smallest elements that are independent from each other and from which all facts are ultimately built.[1] In a predicate logical language, atomic facts correspond to elementary sentences. As we shall see later, in first order predicate logic we can derive from elementary sentences *Pa, Pb, … Qc, … Rab, Rca, … Tadc, …*, quantified formulas 'there exists an *x* s.th. *Px* … there exists an *x* and there exists a *y* s.th. *Rxy*, and so on. (Such inferences are semantically valid, and they can also be derived in the system of natural deduction for predicate logic.) We can apply Quine's criterion to these quantified formulas and get the ontological commitments of these formulas. Because of the connection with the atomic formulas from which the quantified formulas are derived and the correspondence of the former with atomic facts, this gives us a handle on the ontological commitments of a fact-based ontology in terms that are associated with a things-based one.

Of course, information gets lost in the inferences above. Both *Pa* and *Pb* allow us to infer that 'there exists an x s.th. *Px*' so in that sense indeed the 'things-ontology' that we derive from the 'facts-one' contains less information. It does not give us a complete description of the world (but it does not pretend to do so either). When we add identity to our predicate logical languages, we can keep the information: from *Pa* we can then derive 'there exists an x s.th. *Px* and *x* is identical to *a*'. But then we still have the names that Quine argues need to be analysed away using Russellian descriptions.

Group 2:
"Does Quine's criterion decide the issue of universalia? If so, in what way, and why?"

The answer to this question seems to be negative, for the reasons that are given in connection with the 'all groups' question about reductionism, and for reasons analogous to those given in connection with the questions about 'things-ontology versus facts-ontology'. It is not the criterion itself that decides these issues, but the context of application. And the considerations that are relevant there are of an independent nature. In this case it really turns around argument *pro* and *con* the assumption that universals exist. Quine himself takes a pragmatic stance: if the best scientific theory we have about a certain empirical domain 'needs' to quantify over universals, we have to accept their existence. But note that this is only a 'momentary'

[1] In situation theory (cf., Barwise and Perry, *Situations and Attitudes*, MIT Press, 1983)) a similar concept is used, but the requirement of independence is weakened.

commitment: there is no compelling reason to think that we cannot come up with some future theory that will have at least the same explanatory power and that does not need these universals. So the issue really turns around the question whether there are compelling arguments that are *independent* of empirical concerns for the existence of universals. The ontological commitment criterion is neutral with respect to that, so this really remains a purely philosophical issue.

Group 3:
"Does Quine's criterion also apply to language as used in fiction? What would 'existence' in the context of fiction amount to?"

It seems natural to assume that the question of existence in a work of fiction remains restricted to the context of that piece of fiction. It does not make sense to ask whether the ghost that appears to Hamlet really exists outside the scope of Shakespeare's play, just as it does not make sense to ask (or confirm) that Baldy Li has a stepbrother called Song Gang outside the scope of Yu Hua's *Brothers*. The answers ('no' and 'yes', respectively) make sense only in the context of that particular work of fiction. That suggests that we can apply Quine's criterion to determine the ontological commitments of a work of fiction in much the same way as we use it to determine the commitments of a scientific theory, or an everyday piece of discourse: it is the domain that the variables range over that we are committed to as accepting as *bona fide* ontological entities.

That being said, there are some interesting differences between fictional existence and existence in a non-fictional, i.e., everyday or scientific, context. One such difference concerns truth. Ordinarily, we would go with the assumption that everything that we can predicate of an entity is either true or false. When we are discussing our mutual friend '*a*', then the assumption is that every statement we can formulate about *a* is either true or false: *Pa*? true or false; *Rba*? true or false; … Statements that attribute properties to (and claim relations to hold between) everyday objects and the kind of entities that we encounter in the context of scientific theories obey the law of excluded middle. (Here an exception might be needed for the kind of entities we encounter in quantum physics, where, arguably, the law of excluded middle fails also.) But this is not the case in fiction. Obviously, some questions about fictional characters can be considered true or false, as the case may be: Did Hamlet kill Polonius? Yes; Did Song Gang outlive Baldy Li? No. But other questions simply cannot be answered: Was Hamlet right-handed or left-handed? The play does not say, and neither does what it does say imply a positive or a negative answer to this question.

This indicates that the *kind* of entities that which Quine's criterion of ontological commitment tells us we are committed to accept as existing, are of a different nature than those we encounter in everyday contexts or in science. One way of phrasing this is as follows. Following a suggestion by the German philosopher-logician-mathematician Leibniz, we can regard not individuals as basic entities, but properties, and define individuals as maximally consistent sets of properties. I.e., an individual *a* is uniquely defined by the set of properties it has and the fact that it lacks all other properties. Ordinary individuals seem to satisfy this criterion, but fictional entities clearly do not.

Group 4:
"Can you think of contexts in which, pace Quine, reference to 'merely possible' entities makes sense? With or without their being identity criteria for them?"

It seems clear that in our everyday use of language we often talk about entities that are not 'actual' (or 'real', or 'existent', or whatever term we want to use), but 'possible': they do not exist, but we maintain that they could. Does all such use of language need to be avoided?

One case where the answer seems quite clearly 'no' is when we talk about entities in the past: deceased persons, events that took place in the past, and so on. For example, my great grandfather is long deceased, so he does not exist, he is not real. But that does not make a reference we make to him (using a description or a name) meaningless: we can make statements about him that are meaningful, that can be checked for truth and falsity. So, my great grandfather (on my mother's mother side, just to be specific) is clearly a non-existent individual that nevertheless can be named, described, and about whom true and false statements can be made, questions can be asked, and so on.

Now this is a case of a non-existent individual with clear and definite identity criteria: we can give characteristics that, at least in principle, single out a unique individual as my great grandfather on my mother's mother side. And it appears to be crucial for that being the case that my great grandfather (on ….) although non-existent now, once did actually exist. So, although non-existent, he is not 'merely possible'.

For that, let's turn from the past to the future. Can we meaningfully talk about entities that do not exists now, but that might come into existence in the future? It seems we can. For example, we can speculate about the person who is going to be the next president of the USA. We can claim that they are either going to be a Democrat or a Republican, convinced as we are that independent candidates have no chance whatsoever of winning the presidential elections. Such claims, despite their speculative nature, seem to be meaningful: they can be discussed, argued for and against, and so. So, what does the descriptive phrase 'the next president of the USA' denote? If we have a very deterministic conception of time, we can say that it denotes an actual individual, the one who will actually win the next presidential elections. But if our view of time is not completely deterministic and we accept that there might be several candidates but that it is not settled yet who of them will win in the end, then the phrase seems to lack a specific referent. At least, there is no unique individual that we can now identify as what that phrase 'the next president of the USA' refers to. If that is the case, that we need to accept that the phrase refers to a possible individual.

So, what about the identity criteria of this possible individual? Does it have any? Or is 'the next president of the USA' as unspecified as one of Quine's possible men in the doorway? Note that it does not have the kind of identity criteria that we normally associated with identifying actual individuals. (But note also that these vary a lot, with context. Do we identify by name + date of birth? By DNA-fingerprint? By narrative? These are not easy questions to answer, even in the case of ordinary individuals.) However, it is not that *anything* can be the referent of this description: it has to satisfy certain conditions, minimally. Those conditions do not single out a unique individual independent of context, that is true. But that holds for almost all identifications of individuals that we use in ordinary contexts: names are not unique; descriptions usually apply to a number of individuals. And also note that, except in very special circumstances, we usually do not need identity criteria that single out a unique individual.

Also in other contexts, that do not relate to time, it seems we can have meaningful reference to, and discussion of, possible individuals. Suppose we are dealing with a mathematical statement that has not been proven to be true but also not proven to be false. The standard example in discussions like these is Goldbach's Conjecture: 'Every natural number greater than 2 is the sum of two primes.' We can discuss what a proof of the conjecture might look like; what lemmas

might need to be invoked; what concepts are probably going to play a key role; what conclusions might follow from it; and so on. What we are discussing, viz., a proof of the conjecture, either exists or not (depending on whether the conjecture is true or not, and some assumption about provability so Gödel comes in as well …). Moreover, this being a mathematical matter, if the proof exists, it exists necessarily, and if it doesn't it, it cannot exist.

Now is this 'a mere possible individual', one that lacks identity criteria? We do not have sufficient information to actually single out a piece of mathematical reasoning as a proof. If we had, it would not be a conjecture but a theorem. (*Mutatis mutandis* for the negative case in which the conjecture can be proven to be false.) But we probably have enough information to recognize something as a proof once someone is clever enough to come up with one. In that sense we do have 'identity criteria'. But note that we have them *not* because there actually *is* a proof, or merely because there *could be* a proof. For if the conjecture turns out to be false, there could not be such a thing.

These considerations indicate that thinking about possibilities, including possible individuals, might be not the kind of yes/no affair that Quine suggests it is. There is a continuum of cases in which possible entities of various kinds do seem to allow for meaningful discourse about them.